American Sign Language Fingerspelling Recognition

A Comparative Study: Conventional Classifiers vs. Neural Networks

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Problem Statement & Dataset

Motivation: Automated American Sign Language (ASL) recognition systems are essential to bridge the communication gap between the Deaf community and non-signers in daily interactions

Static Example: Letter A

Static Example: Letter B

Evaluated Architectures

Comparative analysis of three distinct paradigms:

- Gabor Filters + Support Vector Machine (SVM) Classifier
- Deep Learning: Custom Convolutional Neural Network
- Transfer Learning: MobileNetV2

Dataset Characteristics

- Target: 24 static ASL letters (A–Y). Note: J and Z are excluded
- Scale: Over 100,000 RGB images (64×64) collected across 5 distinct individuals.

Data pre-processing

Training Set ⟶ Subjects 1, 2, 3, and 4.
Testing Set ⟶ Subject 5.

Before training:

- We resized all images to a fixed size.
- We normalized pixel values between 0 and 1.
- This makes the input consistent for the models.

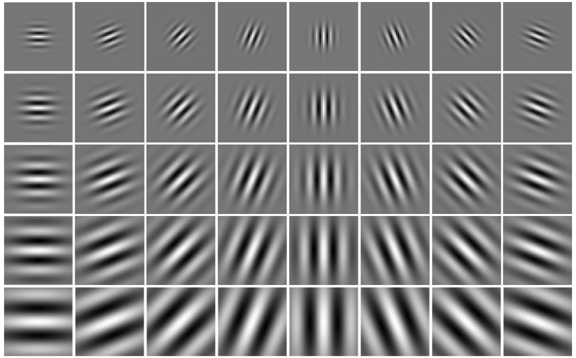
Why this strategy?

To prevent overfitting to specific backgrounds. The model must learn the visual structure of the hand sign, ensuring generalization to unseen individuals.

Figure: Sample from ASL Dataset

Model 1: Gabor Filters & SVM

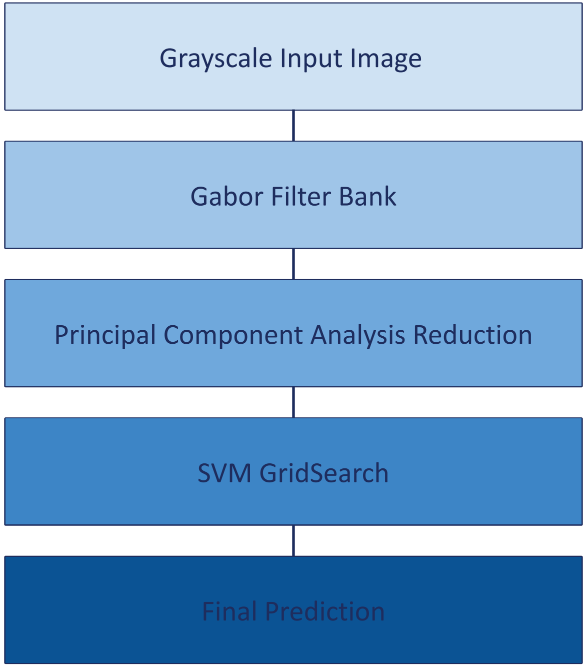
Gabor Bank: 40 filters capturing textures at multiple orientations and frequencies.



filters bank illustration

PCA Reduction: Reducing the Gabor feature vector from 80 to 60 components.

Classification: Support Vector Machine (SVM) with C selected by grid search.

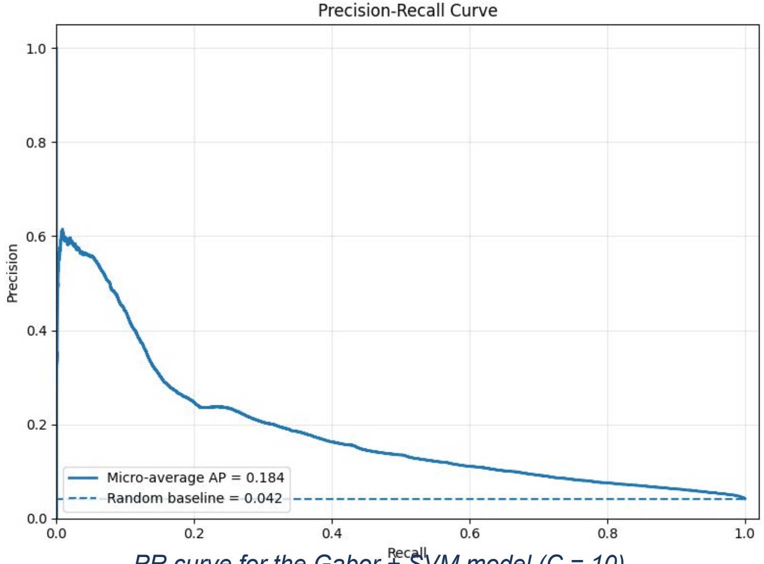


Pipeline of the method

Results: Gabor + SVM Performance

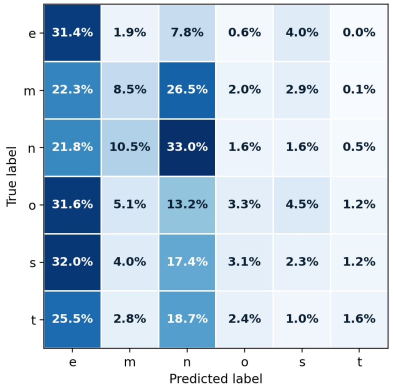
Results of the tests on Subject 5:

Test accuracy	Macro Precision	Macro Recall	Macro F1-score
23%	24%	23%	21%

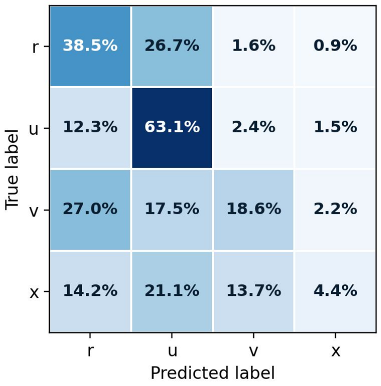


PR curve for the Gabor + SVM model (C = 10), showing performance above the random baseline.

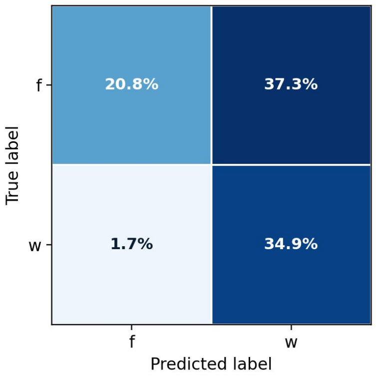
Zoom-ins of the Confusion Matrix (Gabor Features)



Zoom-in 1: compact signs are often confused with e and n .



Zoom-in 2: v and x are often predicted as r or u.

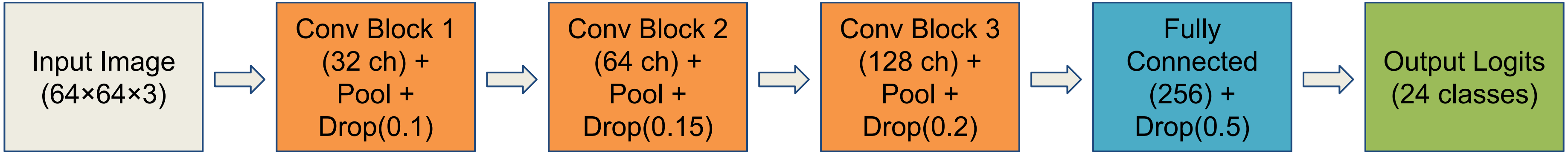


Zoom-in 3: recurrent f / w error.

- Many errors come from visually similar hand shapes.
- Struggles with small finger-position differences.
- Strong isolated f/w confusion.

Model 2: Custom CNN Architecture

Architecture (built from scratch, initialized with random weights) — Input: 64×64



Training setup:

- Optimizer: Adam, $lr = 5 \times 10^{-4}$
- Loss: CrossEntropy + label smoothing 0.1
- 40 epochs, batch size 64

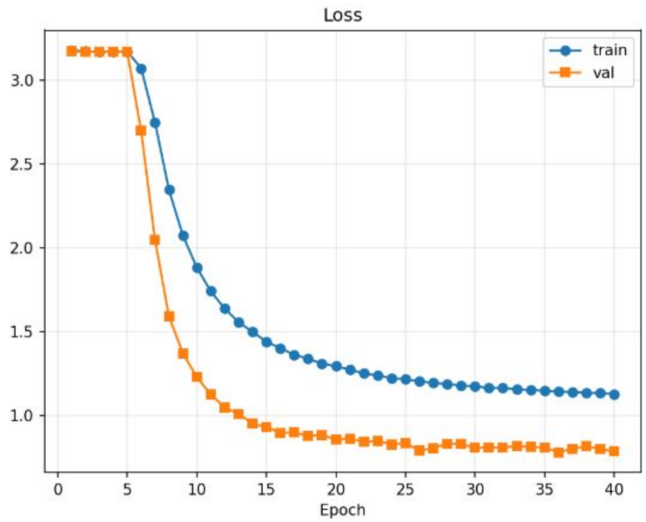
Regularization:

- Dropout2D (0.1 / 0.15 / 0.2) per block
- Strong augmentation: color jitter, random erasing, perspective, blur

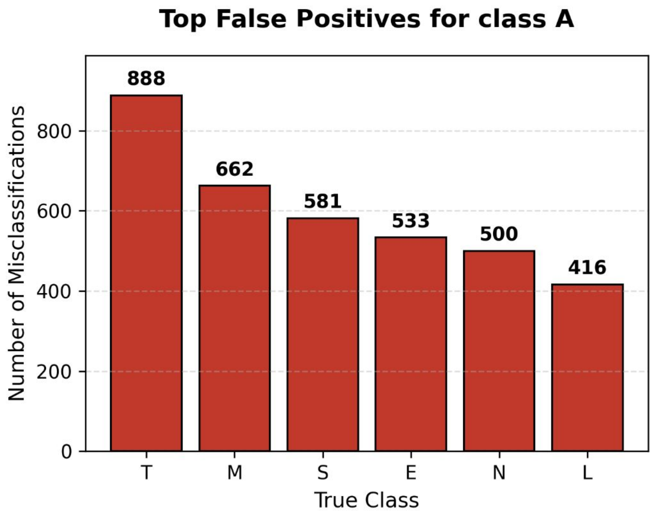
Results: CNN Performance

Test Accuracy: 55.0%

Precision: 0.71 · Recall: 0.55 · F1: 0.57



Training curves (40 epochs)

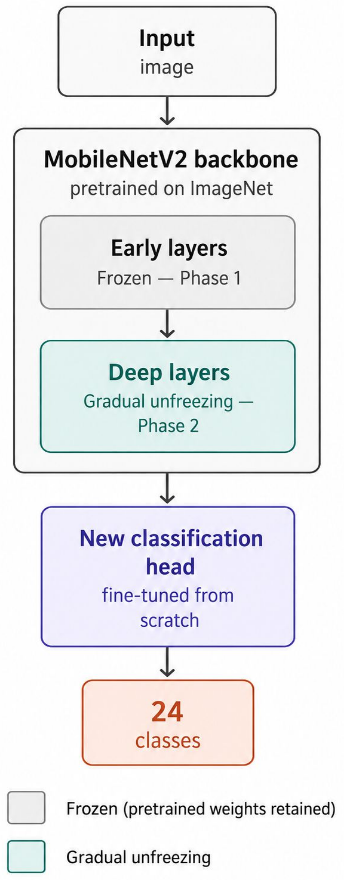


Highest Recall classes : H 98.7%, C 97.1%, A 89.2%, P 84.1%

Most difficult: Closed-fist letters (T, M, S, E, N). Class A acts as a sink (highest False Positive Rate)

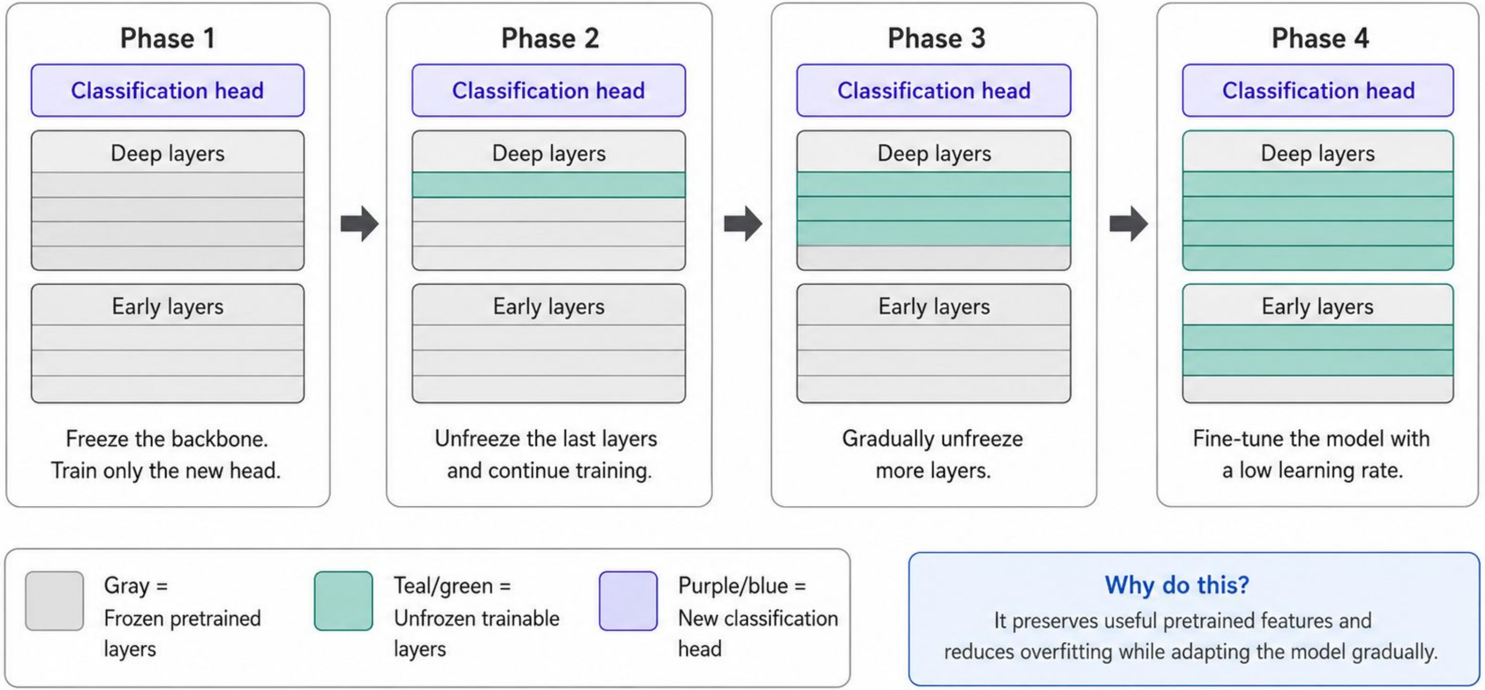
Model 3: Transfer Learning (MobileNetV2)

- Pre-trained MobileNetV2 backbone.
- We replace the last layer for our 24 ASL classes.
- We fine-tune using progressive unfreezing.

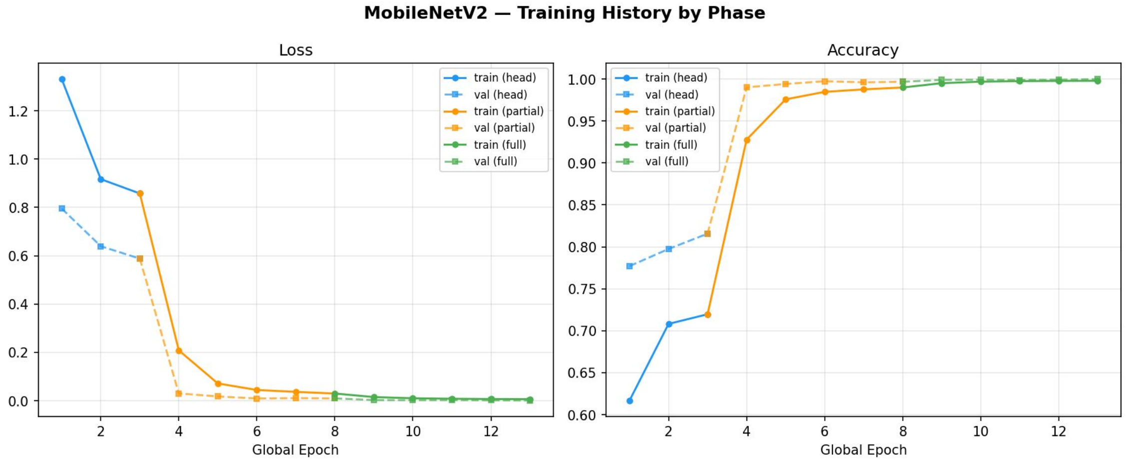


MobileNetV2 was trained on ImageNet. We keep the backbone and replace the last layer for our 24 classes. We fine-tune using progressive unfreezing: first we train only the new head, then we slowly unfreeze deeper layers with a smaller learning rate.

Model 3: Progressive Unfreezing



Results: MobileNetV2 Performance



Global Accuracy:

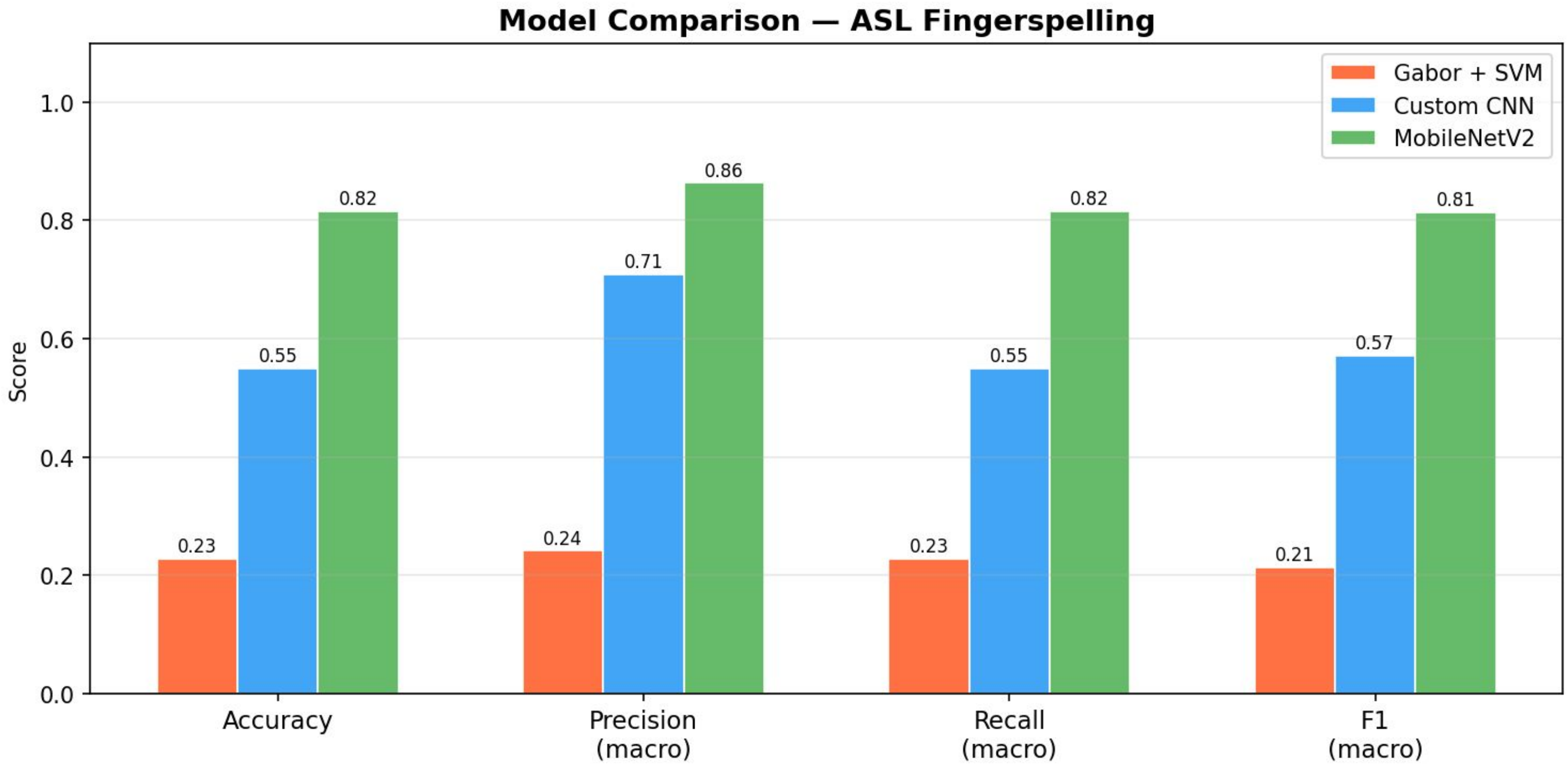
82%

Precision	Recall	F1-score
0.86	0.82	0.81

MobileNetV2 gets 82% accuracy on Subject 5, the best of the three models.

The training curves show 3 phases: first we train only the head, then partial unfreezing, then full unfreezing. Loss drops fast and accuracy reaches nearly 100% on training data.

Final Model Comparison



Conclusion & Future Work

Key Takeaway: MobileNetV2 (82%) clearly outperforms CNN (55%) and Gabor+SVM (23%), confirming the advantage of transfer learning for this task.

Limitations: Sensitive to lighting and complex backgrounds.

Future Work: Use Leave-One-Subject-Out Cross-Validation to get more robust performance estimations.

Questions?